SUPERNOVA REMNANT

## Crab Nebula

TAURUS

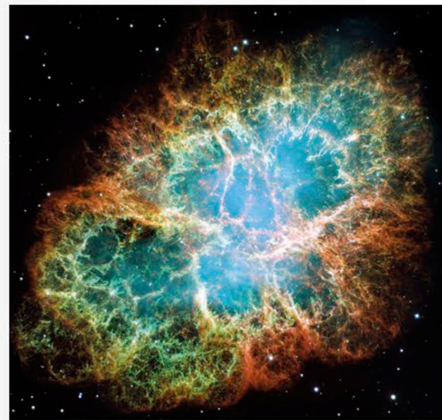
**CATALOG NUMBERS**
M1, NGC 1952

**DISTANCE FROM SUN**
6,500 light-years

**MAGNITUDE** 8.4

In the summer of 1054, during the Sung dynasty, Chinese astronomers recorded that a star, in the present-day constellation Taurus, had suddenly become as bright as the full moon. They described it as a reddish-white "guest star" and observed it over a period of 2 years as it slowly faded. Their records show it was visible in daylight for more than 3 weeks. They had witnessed a supernova, and the stellar material flung off in this cataclysmic explosion now shines as the wispy filaments of the Crab Nebula. This nebula is the very first object, and the only supernova remnant, to be listed by Charles Messier (see p.73) in his famous catalog. The nebula is easily visible in binoculars and small telescopes. It spans a distance of about 10 light-years with a magnitude of between 8 and 9.

The remains of the original star have become a spinning neutron star, a pulsar, rotating at about 30 times per second.

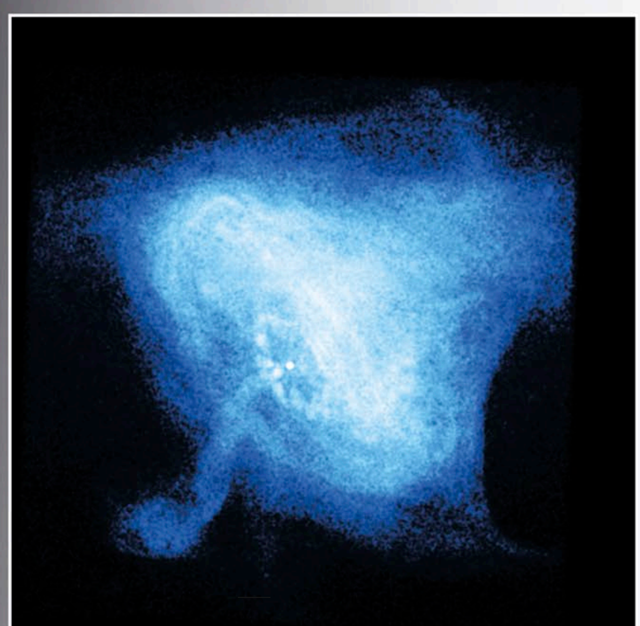


**HUBBLE'S ANALYSIS**
This visible-light mosaic highlights elements expelled in the supernova explosion. Blue indicates neutral oxygen; green indicates ionized sulfur; and red indicates super-hot, doubly ionized oxygen.

**CORE OF THE NEBULA**
This Hubble close-up reveals the Crab Pulsar (upper-right of the two stars at the center), surrounded by the glowing blue haze emitted by electrons traveling close to the speed of light in its powerful magnetic field.

The pulsar (known as PSR 0531+21) is observable optically and in radio, X-ray, and gamma-ray wavelengths because the beams it generates happen to be directed toward Earth during part of its revolution. It was discovered in 1967, but had been known previously as a powerful emitter of radio waves and X-rays. It was the first pulsar to be identified optically and is of 16th magnitude. It is estimated to have a diameter of only about 6 miles (10 km) but a mass greater than the Sun's. Its energy output is more than 750,000 times that of the Sun. Its rotation is decreasing by about 36.4 nanoseconds every day, which means that over 2,500 years from its presently observed state, its rotation period will have doubled (see pp.282–283). The loss of rotational energy is being translated into energy, which is heating the surrounding Crab Nebula.

As the most easily observable supernova remnant, the Crab Nebula has been extensively studied. Detailed observations show that the material within the central portion of the nebula changes within a time scale of only a few weeks. Wispy features, each about a light-year across, have been observed streaming away from the pulsar at half the speed of light. These are created by an equatorial wind emitted by the pulsar (see left). They brighten and then fade as they move away from the pulsar and expand out into the main body of the nebula. The most dynamic feature within the center is the point where one of the polar jets from the pulsar crashes into the surrounding previously ejected material, forming a shock front. The shape and position of this feature have been observed to change over very short time scales.



**PULSAR CLOSE-UP**
This X-ray image of the central region of the Crab Nebula shows its pulsar as a white dot near the center. Jets of matter stream away from the poles of the rapidly rotating pulsar, and energetic particles from its equator plow into the surrounding nebula.